Big Data Frameworks

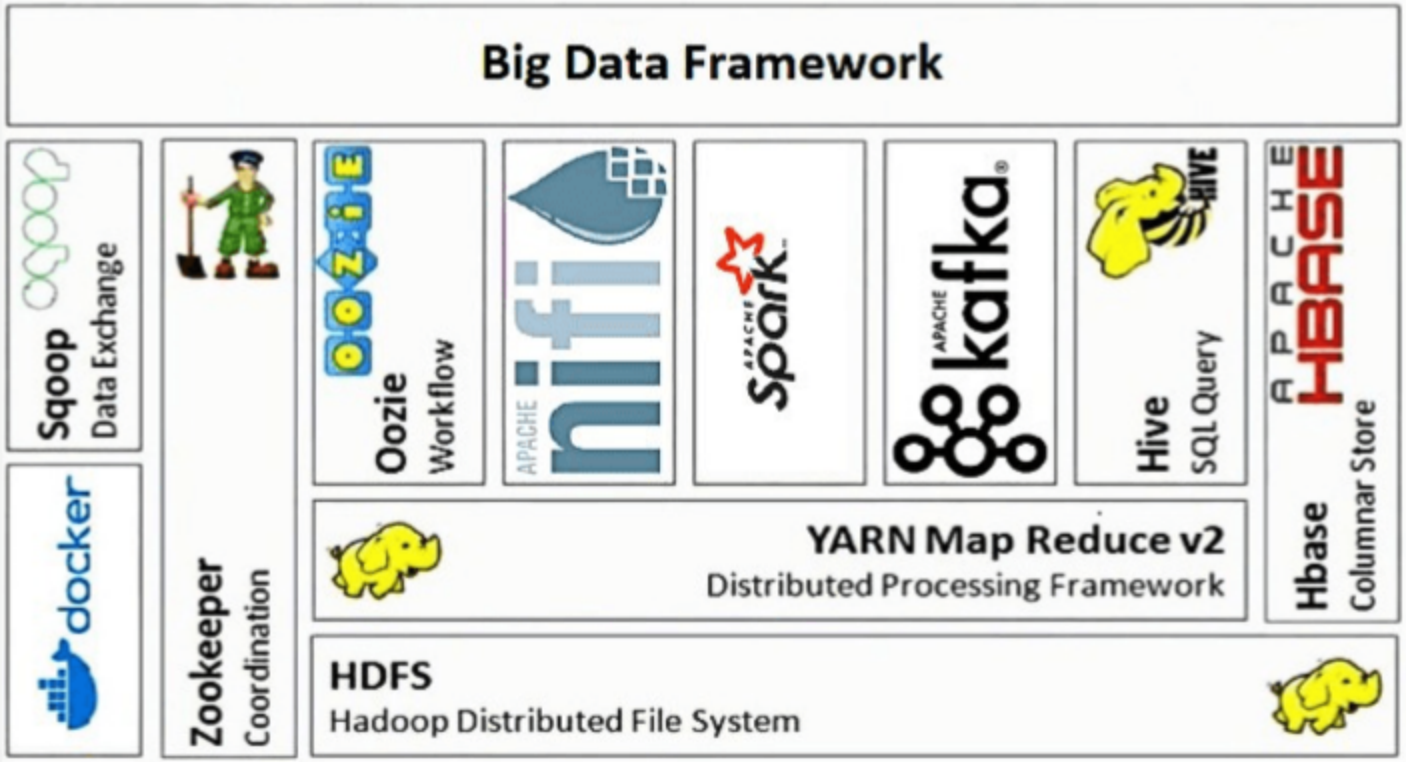
Agenda

- Introduction
- Big Data Frameworks
- Big Data Programming Paradigms
- Big Data Programming Languages
- Application of Big Data

Introduction

Big data frameworks are software tools and platforms designed to process, store, and analyze large and complex data sets, often referred to as big data.

These frameworks are essential for organizations and businesses looking to extract valuable insights, patterns, and knowledge from their data.

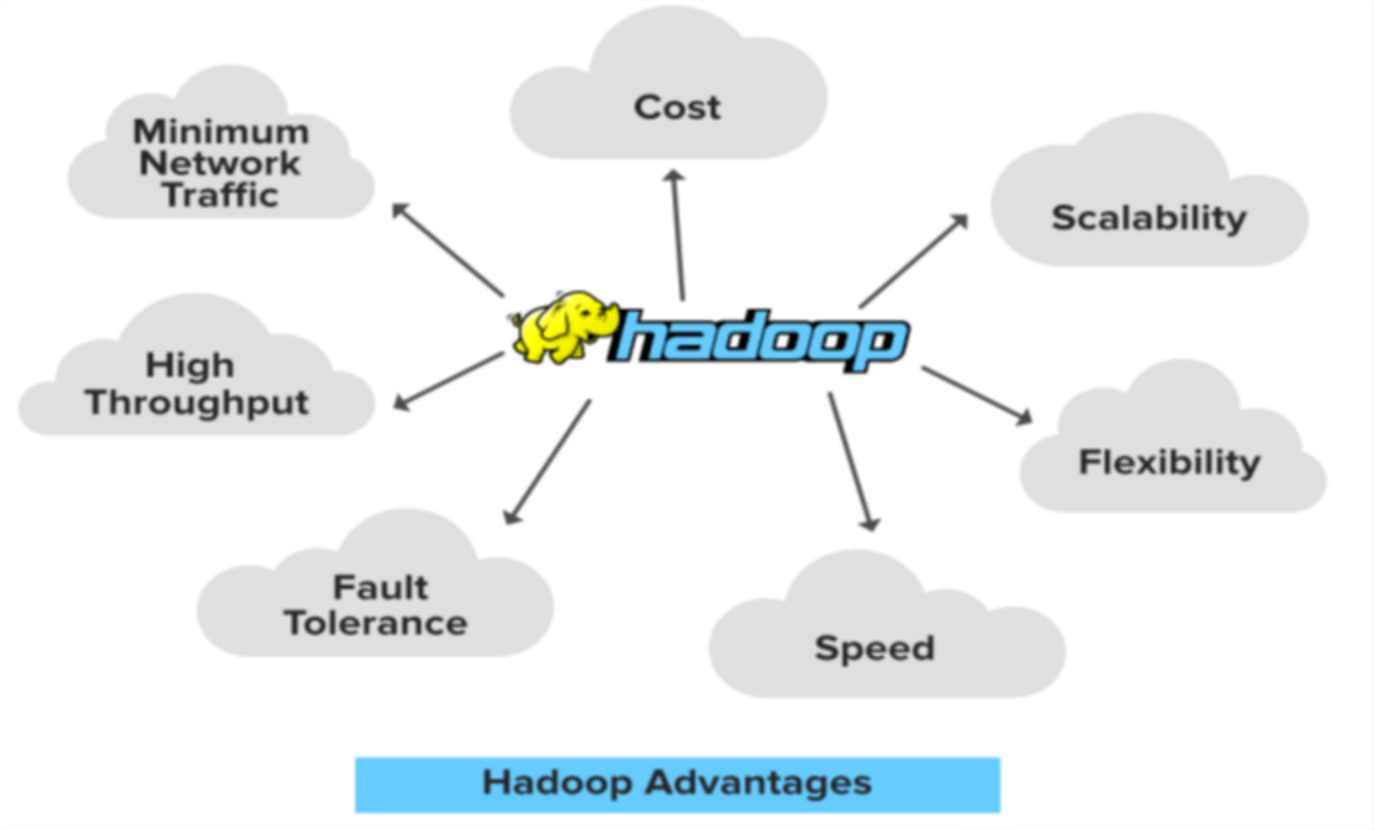


www.object.co.in

Big Data Frameworks

- **Hadoop:**

Hadoop is one of the most well-known and widely used big data frameworks.



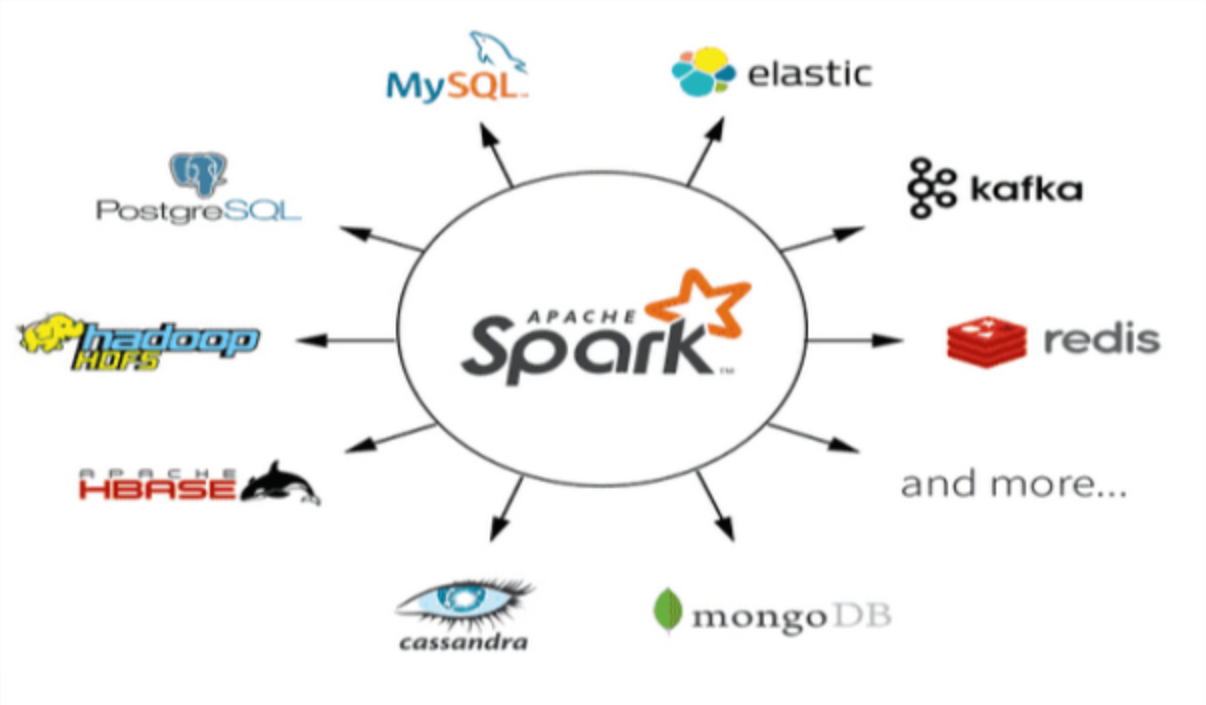
www.object.co.in

Big Data Frameworks

- **Apache Spark:**

Apache Spark is an open-source, fast, and general-purpose cluster computing system.

It provides an in-memory processing engine that can be used for batch processing, real-time data streaming, machine learning, and graph processing.



www.object.co.in

Big Data Frameworks

- **Kafka:**

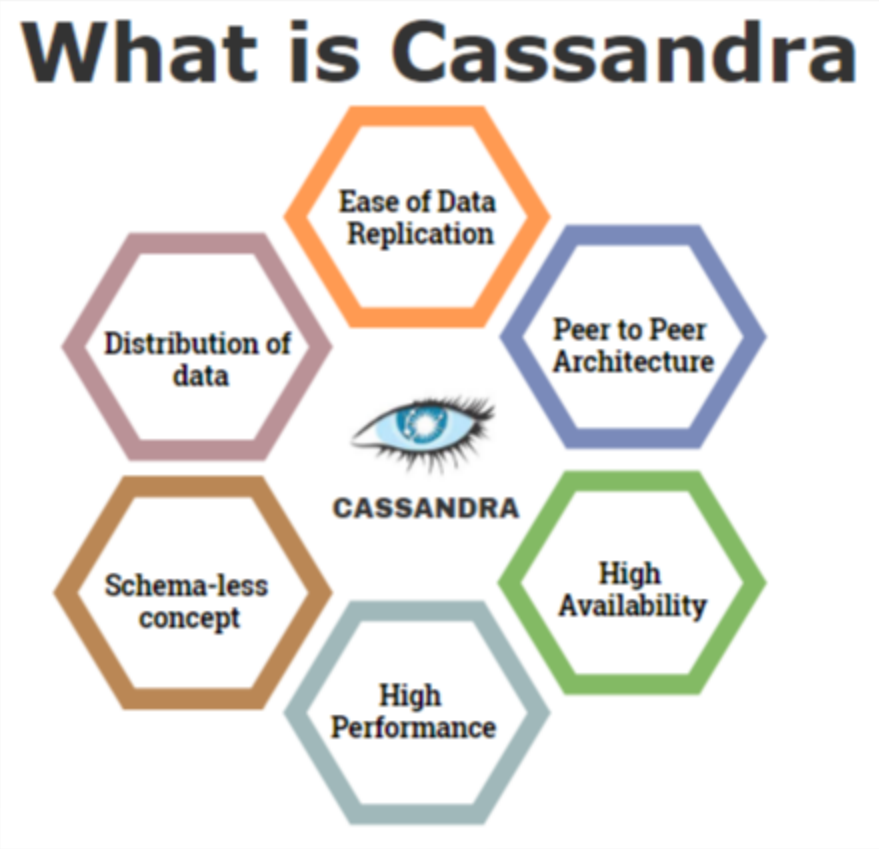
While Kafka is primarily known as a distributed messaging system, it plays a crucial role in big data frameworks by enabling the ingestion and streaming of data at scale.

www.object.co.in

- **Cassandra:**

Apache Cassandra is a highly scalable NoSQL database management system designed to handle large amounts of data across multiple commodity servers.

It is commonly used for distributed and high-availability data storage.



Big Data Programming Paradigms

OBJECT™ TECHNOLOGIES

Big data programming paradigms refer to the approaches, frameworks, and models used to process, analyze, and manipulate large volumes of data, often referred to as big data.

Batch Processing:

MapReduce: MapReduce is a programming model popularized by Google and implemented in systems like Apache Hadoop. It processes large datasets in parallel by dividing the task into two phases: the map phase, where data is filtered and transformed, and the reduce phase, where data is aggregated and analyzed.

Apache Hadoop:

Big Data Programming Paradigms

OBJECT™ TECHNOLOGIES

Stream Processing:

Apache Kafka: Kafka is a distributed streaming platform that allows real-time data processing and event-driven architectures. It can handle high-throughput, fault-tolerant data streams.

Apache Flink: Flink is a stream processing framework that supports event time processing, stateful computations, and exactly-once processing semantics.

Big Data Programming Paradigms

NoSQL Databases:

HBase: HBase is a distributed, scalable, and consistent NoSQL database that stores large volumes of sparse data. It is often used with Hadoop for random read/write operations.

Cassandra: Cassandra is a distributed NoSQL database designed for high write throughput and scalability. It is used for time-series data, event data, and more.

Big Data Programming Paradigms

OBJECT™ TECHNOLOGIES

SQL-Based Processing:

Apache Hive: Hive is a data warehousing and SQL-like query language system built on Hadoop. It allows users to query large datasets using familiar SQL syntax.

Presto: Presto is a distributed SQL query engine designed for interactive queries on large datasets. It supports various data sources, including Hadoop, NoSQL databases, and more.

Big Data Programming Paradigms

OBJECT™ TECHNOLOGIES

Machine Learning and Deep Learning:

TensorFlow: TensorFlow is an open-source machine learning framework developed by Google. It provides tools for building and training machine learning and deep learning models on big data.

PyTorch: PyTorch is another popular open-source deep learning framework that supports distributed training and large-scale model deployment.

Big Data Programming Languages

OBJECT™ TECHNOLOGIES

Java: Java is a widely used language for big data processing, particularly in the Hadoop ecosystem. Apache Hadoop, one of the most popular big data frameworks, is implemented in Java.

Python: Python is a versatile language with a rich ecosystem of libraries and frameworks for big data. It is commonly used for data manipulation, analysis, and machine learning.

Scala: Scala is a language that runs on the Java Virtual Machine (JVM) and is often used in combination with Apache Spark.

R: R is a language specifically designed for statistical analysis and data visualization. It is widely used in data science for tasks like data exploration, statistical modeling, and visualization.

www.object.co.in

Big Data Programming Languages

C/C++: These low-level languages are used in big data systems and libraries to optimize performance-critical components.

Go (Golang): Go is known for its efficiency and is used in some big data projects and distributed systems.

Applications of Big Data

OBJECT™ TECHNOLOGIES

Business Intelligence and Analytics: Big data is used to analyze large datasets to discover patterns, trends, and correlations.

Customer Insights: Companies use big data to gain a better understanding of customer behavior, preferences, and sentiment.

Healthcare: Big data is used for medical research, disease prediction, and improving patient care.

Finance and Fraud Detection: The financial sector uses big data for risk assessment, fraud detection, and algorithmic trading.

Supply Chain Optimization: Big data is used to improve the efficiency of supply chains by analyzing data related to inventory, transportation, and demand forecasting.

www.object.co.in